q-Deformed Numbers and Quasi-Fuchsian Groups

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Near Liberec (2023)

Various quantities have been considered in Analysis which are easy to express but difficult to evaluate. Namely, irrational and transcendental quantities, among which are logarithms, circular arcs, and the lengths of other curves, are frequently expressed by infinite series... Moreover these series are of two kinds... those whose terms are related by addition or subtraction [...and...] those whose terms are related by multiplication.



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Thus, the area of a circle with diameter 1 may be expressed by either method; according to the first, the area of the circle is

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \quad \text{etc. to infinity.}$$

According to the second method, the area equals the expression

$$\frac{2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdot 8 \cdot 8 \cdot 10 \cdot 10}{3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdot 7 \cdot 9 \cdot 9 \cdot 11} \quad \text{etc. to infinity.}$$

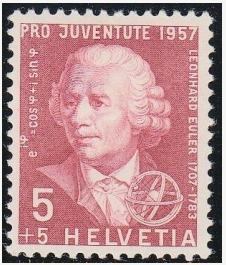
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In addition to the two kinds of series presented above, we consider a third whose terms are connected by continuing division, so that it will he convenient to call these series *continued fractions*. Although this construction is less used than the other two, not only does it exhibit its value just as clearly but it is also very well suited to approximate computation.

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Theorem

Every real number ξ has a continued fraction expansion

$$\xi = [a_0; a_1, a_2, \cdots] := a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\cdots}}}$$

where $a_i \in \mathbb{Z}_{>0}$ if $i > 0$ and $a_0 \in \mathbb{Z}$.

If $\xi \in \mathbb{Q}$ then there are exactly two expansions, both finite length, one odd & one even.

If $\xi \notin \mathbb{Q}$ then there is exactly one expansion, of infinite length.

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E.g. $(1 + \sqrt{5})/2 = [1; 1, 1, 1, 1, \ldots]$.

Corollary

Every positive real number is obtained from 0 *by an essentially unique (infinite) sequence of applications of*

$$A(z) = z + 1 \text{ and } B(z) = \frac{z}{z+1} = \frac{1}{1+1/z}.$$

Every real number can be obtained by taking a positive real number and applying

$$\Phi(z) = -\frac{1}{z}$$

E.g. $A^3(0) = 3$
$BA^3(0) = B(3) = 1/(1+1/3) = 1/(4/3) = 3/4$
$B^2A^3(0) = 1/(1+4/3) = 1/(7/3) = 3/7.$

🤔 $3/7 = [0; 2, 3]$ AND the corresponding word in the semigroup gen. by A, B is B^2A^3.

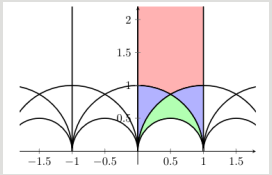
Corollary

Every positive rational number r/s appears as the image of 0 under a word $W_{r/s}$ of

$$\mathrm{PSL}(2,\mathbb{Z}) = \left\langle A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \right\rangle$$

acting on $\mathbb{R}$ by fractional linear transformations:

$$A \cdot z = \frac{1z+1}{0z+1} \quad B \cdot z = \frac{1z+0}{1z+1}.$$



If $r/s = [a_0; a_1, \ldots, a_{2n-1}, a_{2n}]$ then $W_{r/s} = A^{a_0}B^{a_1}\cdots A^{2n-1}B^{a_{2n}}$.

Every negative rational number $-r/s$ is the image of 0 under $\Phi W_{s/r}$ where $\Phi = BA^{-1}B$.

Every real number ξ is the limit of images of 0 under the words $W_{r/s}$ obtained by right-truncating the continued fraction for ξ.

Definition

For $n \in \mathbb{Z}$, the ratio $\dfrac{1-q^n}{1-q}$ is called a q-integer. We denote it by $[n]_q$. Observe that $\lim_{q\to 1}[n]_q = n$.

In general, a *q-analogue* of a mathematical object is an object depending on the variable q that "reduces to" (an admittedly vague term) the original object when we set $q = 1$. To be a "satisfactory" q-analogue more is required, but there is no precise definition of what is meant by "satisfactory".

–R.P. Stanley, *Enumerative Combinatorics*, p. 30, Vol. I, 2nd ed.

Wikipedia, "q-binomial coefficient":

The Gaussian binomial coefficients are defined by:[1]

$$\binom{m}{r}_q = \frac{(1-q^m)(1-q^{m-1})\cdots(1-q^{m-r+1})}{(1-q)(1-q^2)\cdots(1-q^r)}$$

where m and r are non-negative integers. If $r > m$, this evaluates to 0. For $r = 0$, the value is 1 since both the numerator and denominator are empty products.

Although the formula at first appears to be a rational function, it actually is a polynomial, because the division is exact in $\mathbf{Z}[q]$.

All of the factors in numerator and denominator are divisible by $1 - q$, and the quotient is the q-number:

$$[k]_q = \sum_{0\le i<k} q^i = 1 + q + q^2 + \cdots + q^{k-1} = \begin{cases} \frac{1-q^k}{1-q} & \text{for} \quad q \neq 1 \\ k & \text{for} \quad q = 1 \end{cases},$$

Dividing out these factors gives the equivalent formula

$$\binom{m}{r}_q = \frac{[m]_q[m-1]_q\cdots[m-r+1]_q}{[1]_q[2]_q\cdots[r]_q} \quad (r \le m).$$

Theorem (Morier-Genoud & Ovsienko, 2020)

There exists a unique map

$$[\cdot]_q : \mathbb{Q} \cup \{\infty\} \to \mathbb{Z}[[q]]$$

satisfying $[0]_q = 0$, $[x+1]_q = q[x]_q + 1$, *and* $[-\frac{1}{x}]_q = -\frac{1}{q[x]_q}$.

These recurrence relations can be used to define the q-deformed modular group

$$\mathrm{PSL}(2,\mathbb{Z})_q = \left\langle A = q^{-1/2}\begin{bmatrix} q & 1 \\ 0 & 1 \end{bmatrix}, \quad \Phi = q^{-1/2}\begin{bmatrix} 0 & -1 \\ q & 0 \end{bmatrix} \right\rangle.$$

When $q = 1$, $\mathrm{PSL}(2,\mathbb{Z})_q = \mathrm{PSL}(2,\mathbb{Z})$. Also, $B = \Phi A^{-1}\Phi$ can be used to define $W_{r/s}$.

The Laurent series for $[r/s]_q$ satisfy sufficiently nice convergence properties for $[\cdot]_q$ to extend to all real numbers: if $\alpha_i \to \xi$ is a sequence of continued fraction approximates then the series for $[\alpha_i]_q$ converge to a Laurent series for $[\xi]_q$.

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When $q = 1$, $\mathrm{PSL}(2,\mathbb{Z})_q = \mathrm{PSL}(2,\mathbb{Z})$. The image of 0 under $W_{r/s} \in \mathrm{PSL}(2,\mathbb{Z})_q$ is the Laurent series $[r/s]_q$.

Observation (Morier-Genoud, Ovsienko, Veselov 2024): The convergence of these Laurent series for particular q depends strongly on whether or not $\mathrm{PSL}(2,\mathbb{Z})_q$ is faithful.

Conjecture: $\mathrm{PSL}(2,\mathbb{Z})_q$ is faithful in the complement of $\frac{3-\sqrt{5}}{2} < |q| < \frac{3+\sqrt{5}}{2}$.

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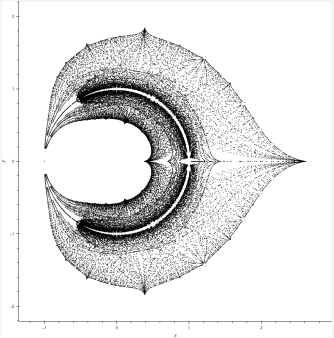
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Theorem (E., Gong, Martin, Schillewart 2024): $\mathrm{PSL}(2,\mathbb{Z})_q$ is faithful when $-q$ lies in one of the two unshaded regions of the figure (stronger than conjecture).

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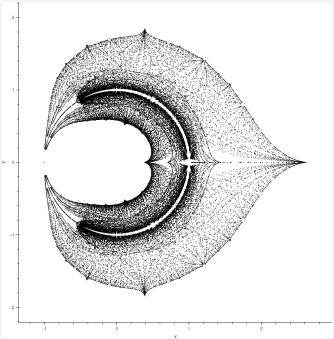
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Recently Etingof–Ovsienko (2026) generalised this radius of convergence statement to all $\xi \in \mathbb{R}$ (different techniques).



What is going on?

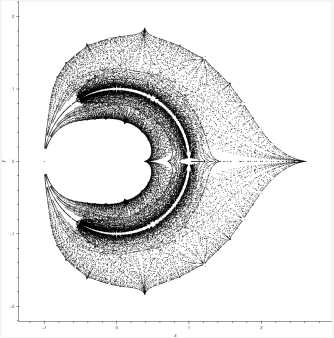
The group $\mathrm{PSL}(2,\mathbb{Z})$ acts discontinuously on $\mathbb{H}^2 \subset \mathbb{C}$ & acts ergodically on $\mathbb{R}$.

Taking

$$\mathrm{PSL}(2,\mathbb{Z})_q = \left\langle A = q^{-1/2}\begin{bmatrix} q & 1 \\ 0 & 1 \end{bmatrix}, \quad \Phi = q^{-1/2}\begin{bmatrix} 0 & -1 \\ q & 0 \end{bmatrix} \right\rangle.$$

and deforming q away from 1, there is an open region $\mathcal{Q}$ where the groups continue to be discrete.

The groups $\mathrm{PSL}(2,\mathbb{Z})_q$ for $q \in \mathcal{Q}$ still act discontinuously on the complement of a Jordan curve $C_q \subset \mathbb{C}$, and act ergodically on a subset $\Lambda_q \subset C_q$.



unshaded region = $\mathcal{Q}$ = connected set of $-q$ such that $\mathrm{PSL}(2,\mathbb{Z})_q$ is discrete

Observation (Bapat, Becker, and Licata 2023). There is a second plausible choice of $[r/s]_q$ for $q \in \mathbb{Q}$ which arises from 'a different orbit of B_3 on the boundary of an embedding of the stability space of the 2-Calabi–Yau category of the A_2 quiver into an infinite-dimensional projective space.'

Theorem (E., 2026)

For $q \in \mathcal{Q}$, the set

$$\{\textit{both choices of } [\xi]_q : \xi \in \mathbb{R}\}$$

is canonically identified with the set on which $\mathrm{PSL}(2,\mathbb{Z})_q$ *acts ergodically.*

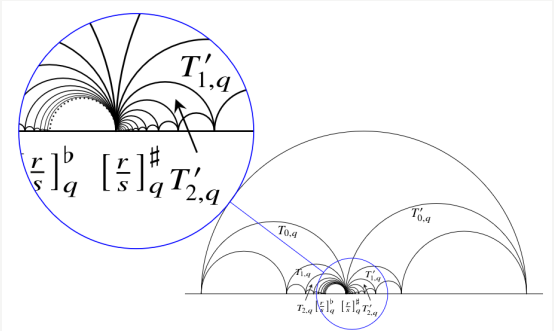


Figure 5. *The sequence of triangles $T_{n,q}$ converging to the q-deformed rational curve of r/s, shown as a dotted line.*

Bapat, Becker, Licata, "q-deformed rational numbers and the 2-Calabi–Yau category of type A_2", Forum Math. Sigma 2023.

Since $\mathrm{PSL}(2,\mathbb{Z})_q$ is quasi-Fuchsian we can rely on classical theory of Ahlfors–Bers–Maskit in complex analysis to prove in a few lines:

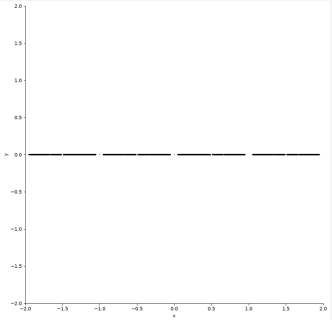
Corollary

For $q \in \mathcal{Q}$:

- *The set of realisations of the q-deformed real numbers is a Cantor set that lies on a Jordan curve.*
- *For $q \in (-1, 0)$, the closure of the q-deformed Farey triangulation is a topological disc.*
  - *It's basically the universal cover of $\partial CC(\mathbb{H}^3 / \mathrm{PSL}(2,\mathbb{Z}))$.*
  - *This resolves a conjecture of Bapat–Becker–Licata.*
- *As $q \to 0$, all q-real numbers limit onto $\{0, 1, \infty\}$.*



Lars Ahlfors
Not. AMS. 1998



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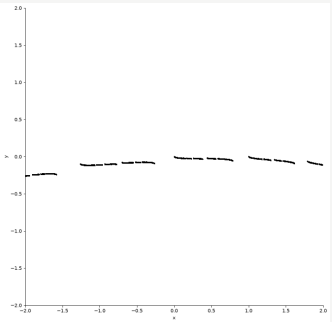
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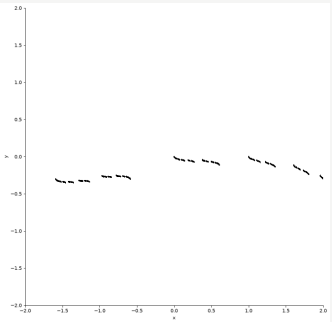
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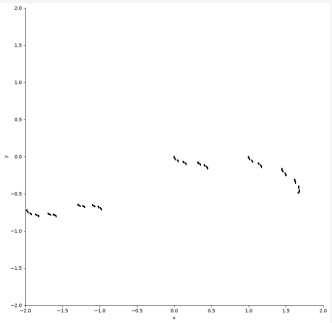
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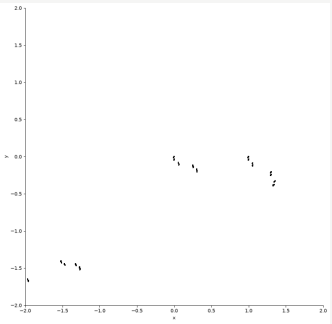
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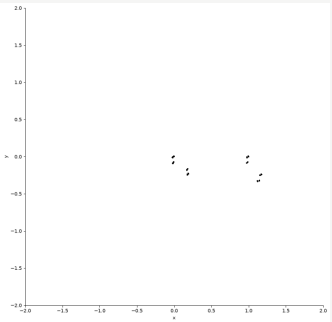
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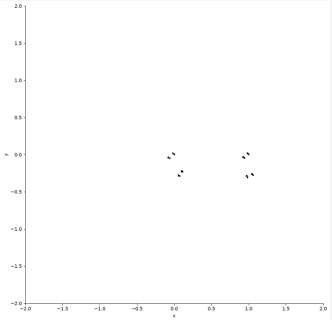
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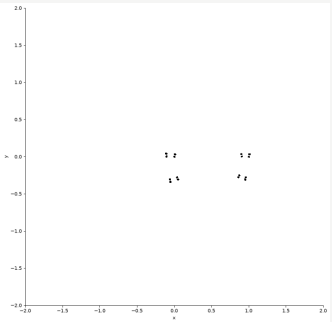
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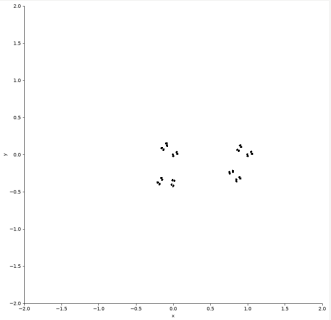
Corollary

For $q \in \mathcal{Q}$:

- *The set of realisations of the q-deformed real numbers is a Cantor set that lies on a Jordan curve.*
- *For $q \in (-1, 0)$, the closure of the q-deformed Farey triangulation is a topological disc.*
  - *It's basically the universal cover of $\partial CC(\mathbb{H}^3 / \mathrm{PSL}(2,\mathbb{Z}))$.*
  - *This resolves a conjecture of Bapat–Becker–Licata.*
- *As $q \to 0$, all q-real numbers limit onto $\{0, 1, \infty\}$.*



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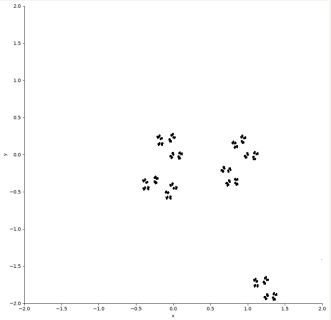
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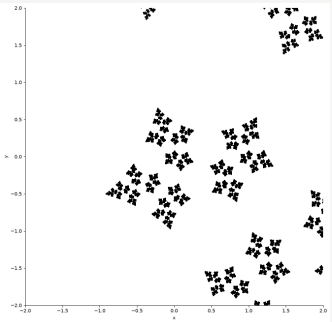
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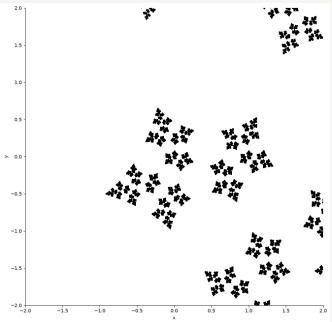
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Combinatorial techniques rely on hard technical arguments, our proofs are entirely conceptual.



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Selected reading

- A. Elzenaar, “Quasi-Fuchsian groups and complex realisations of q-deformed real numbers”.
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- A. Elzenaar, J. Gong, G. Martin, & S. Schillewaert, “Bounding deformation spaces of Kleinian groups with two generators”.
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- S. Morier-Genoud & V. Ovsienko. “q-deformed rationals and q-continued fractions”.
  Forum Math. Sigma (2020). doi: 10.1017/fms.2020.9
- S. Morier-Genoud & V. Ovsienko. “On q-deformed real numbers”
  Exp. Math. (2022). doi: 10.1080/10586458.2019.1671922
- A. Bapat, L. Becker, A. Licata. “q-deformed rational numbers and the 2-Calabi-Yau category of type A_2”
  Forum Math. Sigma (2023). doi: 10.1017/fms.2023.32
- P. Jouteur, O. Paris-Romaskevich & A. Thomas. “Plane geometry of q-rationals and Springborn operations”
  arXiv: 2603.04295

